

Overview

This lecture introduces the **basics of computing**, covering **computer systems**, **data processing**, and **applications of computers in society**. It also explores **hardware and software** components and their roles in the **information processing cycle**.

Topics

- | | |
|---------------------------------|-------------------------------------|
| 1. Computer Definitions | 4. Computer Applications in Society |
| 2. Basics of Computers | 5. Categories of Computers |
| 3. Information Processing Cycle | |

Objectives / Outcome

1. Understand the fundamental concepts of **computing**.
2. Learn about **computer hardware** and **software**.
3. Gain insights into **data representation** and **processing**.
4. Explore the basics of **networks** and **cybersecurity**.

Computer Definition

Computer

- An **electronic device that processes data**, converting it into information that is useful to people.
- It performs four basic operations:
 - **Input**
 - **Processing**
 - **Output**
 - **Storage**

Computer Literacy

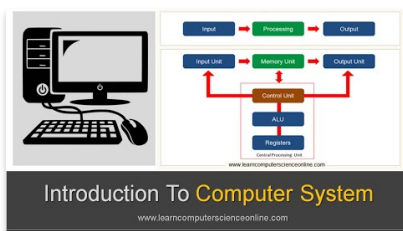
- The **ability to use computers effectively and efficiently**.
- It involves understanding **computer basics**, **software applications**, and the **internet**.

Importance

- **Enhances productivity and efficiency**.
- **Essential for most modern job roles**.
- **Facilitates access to information and digital communication**.

Introduction to Computer System

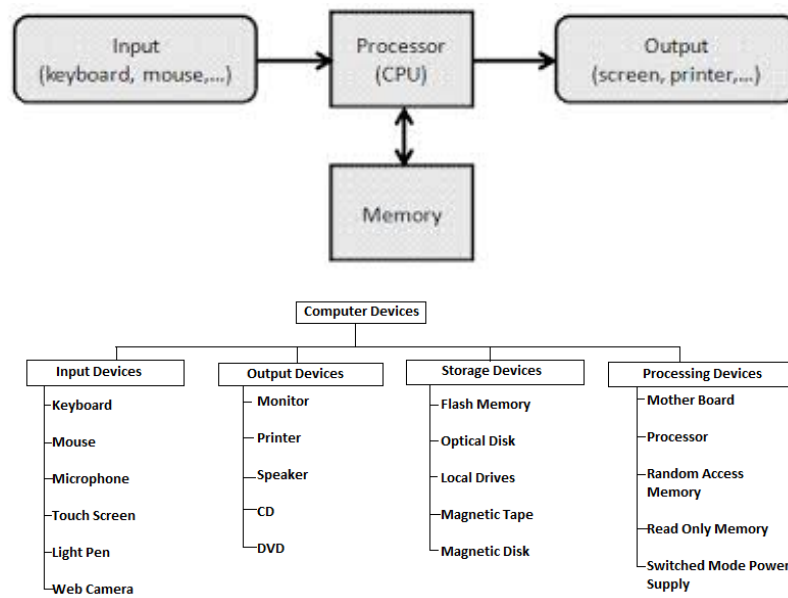
Video



Basics of Computers

Components

Hardware and software components make up a computer system. Key images for reference:



Hardware

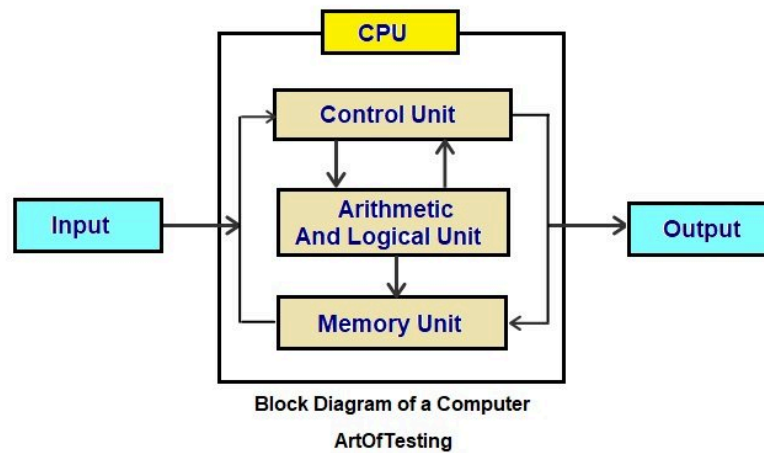
- **Physical parts of a computer system** such as CPU, memory, storage devices, and input/output devices.

Software

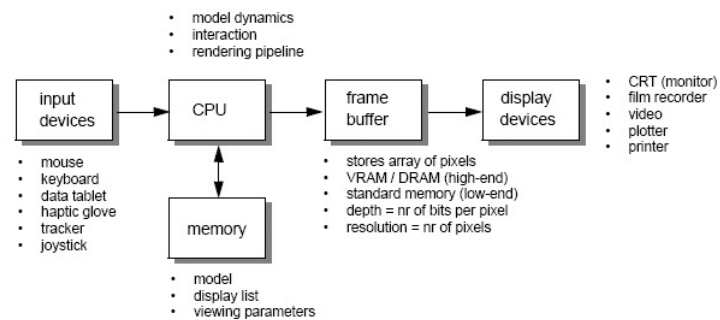
- **Programs and operating systems** that run on computers.

Operations

- **Input:** Data entered into the computer (e.g., keyboard, mouse).
- **Processing:** CPU processes the data.
- **Output:** Processed data is output (e.g., monitor, printer).
- **Storage:** Data is saved for future use (e.g., hard drive, SSD).



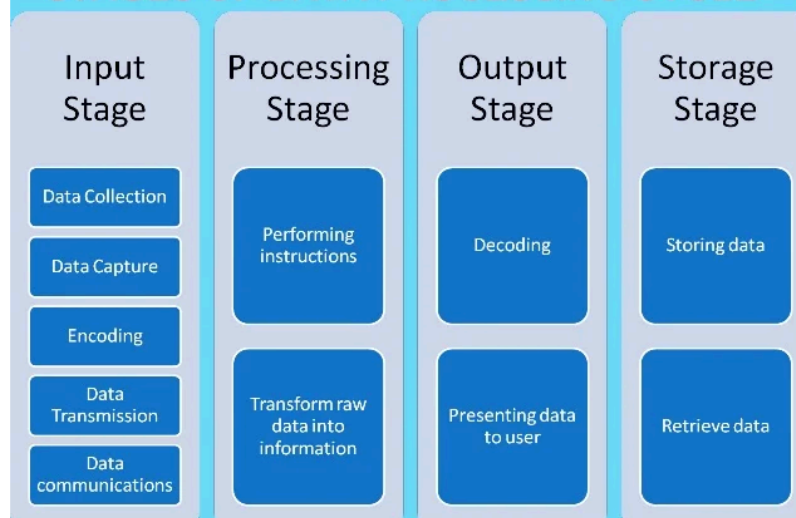
Types of Components in Hardware.



Stages

- **Input:**
 - Entering data into the computer system.
 - Example: Typing on a keyboard.
- **Processing:**
 - Performing operations on the data.
 - Example: Calculations by the CPU.
- **Output:**
 - Displaying the processed data.
 - Example: Showing results on a monitor.
- **Storage:**
 - Saving data for future use.
 - Example: Saving a file on a hard drive.

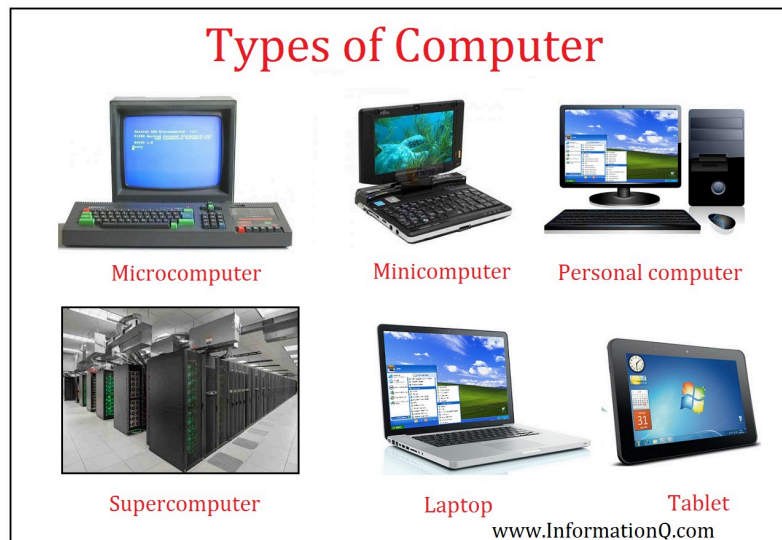
STAGES OF DATA PROCESSING CYCLE



Categories of Computers

Types

- **Supercomputers:** Extremely fast computers used for complex scientific calculations.
 - Example: Weather forecasting systems.
- **Mainframe Computers:** Large, powerful systems used in businesses for bulk data processing.
 - Example: Banking systems.
- **Minicomputers:** Mid-sized computers used in small to medium-sized businesses.
 - Example: Manufacturing control systems.
- **Microcomputers:** Personal computers, including desktops, laptops, and tablets.
 - Example: Home and office computers.



Summary

In this lecture, we have covered the **basic concepts of computing**, including the **definition and importance of computers**, **computer literacy**, the **information processing cycle**, and various **applications of computers in society**. We have also explored the **different categories of computers**, highlighting their specific uses and characteristics.

References

- Brookshear, J. Glenn. "Computer Science: An Overview."
- Patt, Yale N., and Sanjay J. Patel. "Introduction to Computing Systems."
- <https://informationq.com/what-is-computer-types-of-computer/>